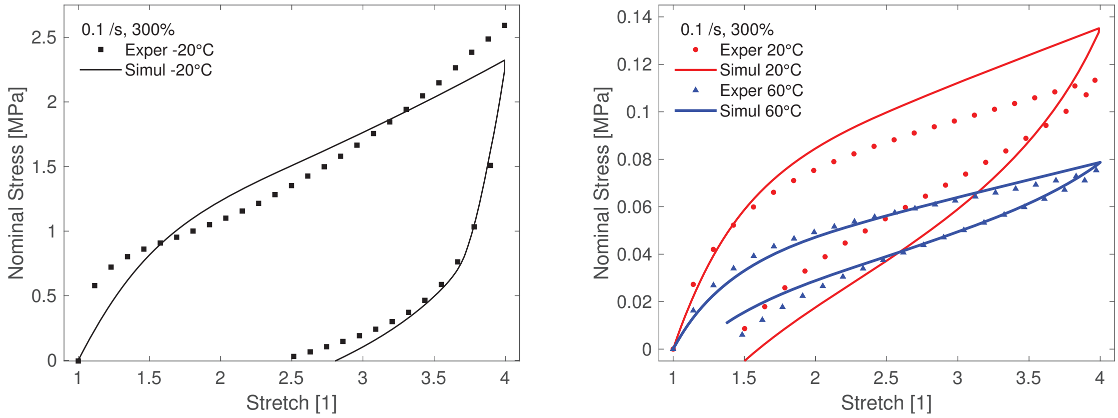
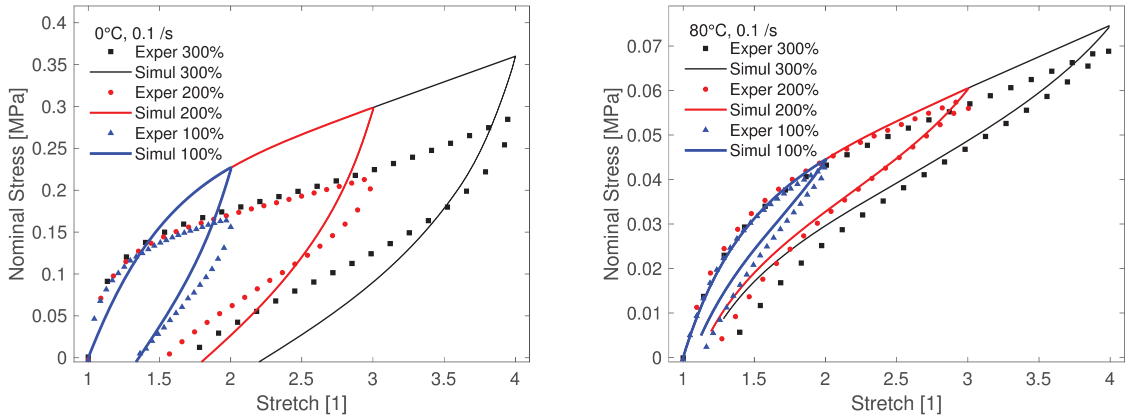


(a) (b)

Figure 17: Thermo-viscoelastic parameter identification taking into account of a wide range of temperature data from -20°C to 60°C with a strain of 300% and a strain rate of 0.1 /s: (a) -20°C, (b) 20°C and 60°C.



(a) (b)

Figure 18: Thermo-viscoelastic model validation from 100% to 300% strains at 0.1 /s strain rate: (a) 0°C, (b) 80°C.

5. Conclusion

In this contribution, a detailed thermo-viscoelastic experimental characterization of the widely-used VHB polymer is presented. In order to quantify viscoelastic behaviour of the polymer, all classical experimentations, i.e., quasi-static tests at very low strain rates, single-step and multi-step relaxation tests, and loading-unloading cyclic tests are conducted at different strain rates and at different temperatures. Relevant viscoelastic experiments reveal a pronounced influence of the temperature on the VHB polymer where an increasing positive temperature above the reference temperature (room temperature) softens the material mechanically and vice-versa. A thermodynamically consistent phenomenologically motivated thermo-viscoelastic material model at finite strain is proposed where the time-invariant hyperelastic part is derived from a Carrol type energy function. At first, we identify hyperelastic material parameters of the Carrol type model using the quasi-static experimental data. For viscoelastic parameter identification, a simultane-